



An example showcasing the features of the msheet package

1 The msheet class (msheet.cls)

The icon on the left hand side of the title box is set according to the chosen document type. Available options are *worksheet*, *factsheet*, *groupwork*, *schedulework* and *experiment*. This document is of type *factsheet*.

We can force a rounded title box using the option *rounded*. The background color of the title box might be chosen as well: *cyan*, *grey*, *white* and *yellow*.

The line spacing is set with *singlespacing*, *onehalfspacing* (used here) and *doublespacing*.

All of these can also be set with `\msheetsetup`, which works in the preamble and in the document body. A license and an attribution are given with `\license` and `\attribution` (see the footer); both are optional.

2 The msheet style (msheet.sty)

2.1 Headings

`\mchapter` and `\mtestpart` typeset centred coloured headings:

Ein Kapitel

Ein Teil

2.2 Exercise headings

Aufgabe 1 *A subsection like task environment.*

Compute the average velocity between $t = 5$ and $t = 8$.

a) $f(t) = 6t$

b) $f(t) = 6t + 2$

c) $f(t) = \frac{1}{2}at^2$

d) $f(t) = t - t^2$

e) $f(t) = 6$

Aufgabe *And the same as starred version, without a number.*

Compute the average velocity between $t = 5$ and $t = 8$.

Aufgabe 2 *A run-in heading: the text continues here.* The body follows on the same line as the heading.

2.3 Lists

- a) first
- b) second
 - (1) nested, numbered (1)
 - (2) and a second one
- c) third

»»» Bitte wenden! »»»

2.4 Inline markers

Hints in the running text use `\mtag`, with `\mtip` and `\msolution` as the two shorthands: `\mtip 1` `Lösung 1` `beliebige Farbe`. `\mibnp` puts a „Bitte wenden!“ marker at the foot of the page and breaks it.

2.5 Boxes

Each box is named after what goes in it; the starred form drops the title, and the optional argument replaces the default one.

Satz 1 – l'Hôpitals rule

Suppose $f(x)$ and $g(x)$ both approach zero as $x \rightarrow a$. Then $\frac{f(x)}{g(x)}$ approaches the same limit as $\frac{f'(x)}{g'(x)}$, if that second limit exists:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

The starred form, for a statement that needs no heading.

Versuch 1 – The gravitational force

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Definition

The default title of `mexample` is „Beispiel“; here it has been replaced.

Information

The default title of `minfo` is „Information“.

And without a title.

A free title in an arbitrary colour

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

A neutral frame with no title at all — the workhorse box.

2.6 Scores

`\mscore` puts the score of an exercise into the margin and adds it to a running total; `\mtotalscore` prints that total, and `\mresetscore` starts a new count.

Aufgabe 3 *Scored exercise*

Compute $\int_0^1 x^2 dx$.

4 BE

Aufgabe 4 *Another scored exercise*

Compute $\int_0^1 x^3 dx$.

2.5 BE

6.5 BE

Viel Erfolg!